

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)



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SEMESTER 1 2025/2026

WEEK 7: PLC INTERFACING

SECTION 2

GROUP 17

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Date of Experiment: Monday, 17 November 2025

Date of Submission: Monday, 24 November 2025

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ABSTRACT

This laboratory experiment focused on implementing an industrial Start-Stop Control Circuit using a Programmable Logic Controller (PLC) approach integrated with an Arduino microcontroller. The primary goal was to design, simulate, and deploy a latching control circuit, a fundamental concept in industrial automation, onto a low-cost embedded platform. We utilized the OpenPLC Editor, an IEC 61131-3 compliant software, to develop the control logic in Ladder Diagram (LD).

The experiment involved creating a specific LD, defining variables, and associating them with the Arduino Uno's physical I/O pins, specifically using two push buttons for Start and Stop, and a Red LED as the output. After successful simulation and compilation within the OpenPLC Editor, the program was transferred to the Arduino board via the COM port. The physical circuit was constructed on a breadboard to test the system's functionality: ensuring the LED remains ON after a momentary Start press (latching) and turns OFF only after the Stop press (unlatching). The results demonstrated the practical application of PLC principles for controlling electromechanical processes using accessible hardware.

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INTRODUCTION

This laboratory experiment aims to bridge the gap between industrial automation control and accessible embedded hardware by implementing the fundamental control logic of a Start-Stop Control Circuit using a Programmable Logic Controller (PLC) approach integrated with an Arduino microcontroller. The primary purpose of this exercise is to gain hands-on experience in the design, simulation, and deployment of this critical control program. The key objectives include mastering the use of the OpenPLC Editor (an IEC 61131-3 compliant software) to create the latching logic in the Ladder Diagram (LD) language, associating the logical variables within the LD with the physical Input/Output (I/O) pins of the Arduino board (specifically the push buttons and LED), and successfully transferring the compiled program to the Arduino to verify the circuit's functionality.

A PLC is a specialized industrial computer designed for reliable control. The Ladder Diagram (LD) is the graphical programming language used, mimicking traditional relay logic. The core control function studied is the Start-Stop Control Circuit, which relies on a latching mechanism. This mechanism uses an internal coil (Latch) to seal its own input path, ensuring that a momentary press of the Start button maintains the output (the LED) ON. The circuit remains latched until the Stop button, wired as a Normally Closed (NC) contact, is pressed to interrupt the power flow and unlatch the circuit. OpenPLC Editor is the software used to create this code, and a crucial step is physical addressing, where logical program variables are mapped to the Arduino's physical pins using syntax like %QX0.0 (Digital Output) or %IX0.0 (Digital Input).

Based on the principles of Ladder Diagram logic and the expected behavior of the control circuit, a clear hypothesis is established. It is hypothesized that the developed LD program will successfully control the connected hardware according to the specified logic. Specifically, a momentary press of the UP_Start push button will turn the LED ON (latch the circuit), and the LED will remain ON until the DOWN_Stop push button is pressed, which will turn the LED OFF (unlatch the circuit).

MATERIALS AND EQUIPMENTS

Hardware and Components

- Microcontroller Board: Arduino Mega Board
- Input Devices: 2 Push Button Switches
- Output Device: 1 LED (Red LED LED1)
- Resistors:
 1. 2 x 10k ohm Resistors (R1, R2) for pull-down inputs
 2. 1 x 560 ohm (R3) for the LED current limiting
- Interfacing: Jumper Wires
- Prototyping: Breadboard

Software and Equipment

- PC/Laptop: Used for programming and power supply
- PLC Editor Software: OpenPLC Editor (IEC 61131-3 compliant PC editor)
- USB Cable: To connect the Arduino Mega board to the PC/laptop

EXPERIMENTAL SETUP

Electronic Circuit Diagram

The control circuit is set up according to the **Start-Stop Control Circuit** schematic.

- Power: The circuit is powered by the Arduino's +5V and GND pins.
- Inputs (Push Buttons):
 - PB1 (Stop Button): Connected to Arduino Pin A9 (mapped to %IX0.1). A 10k ohm pull-down resistor (R1) is used.
 - PB2 (Start Button): Connected to Arduino Pin A8 (mapped to %IX0.0). A 10k ohm pull-down resistor (R2) is used.
- Output (LED):
 - LED1: Connected to Arduino Pin 14 (mapped to %QX0.0) through a 560 ohm current-limiting resistor (R3).

Physical Breadboard Setup

The components are mounted on a breadboard. The push buttons are connected in parallel to the +5V rail and to their respective input pins (Pin A8 and Pin A9) via the pull-down resistors to GND. The LED and its current-limiting resistor are connected between Pin 14 and GND.

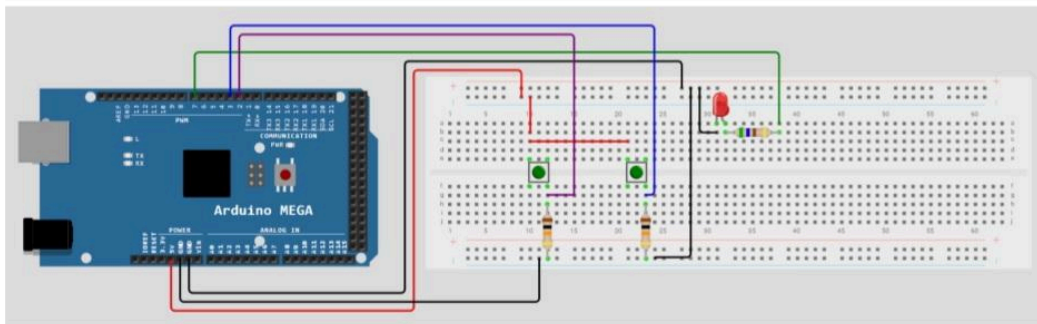


figure 1.0 schematic diagram

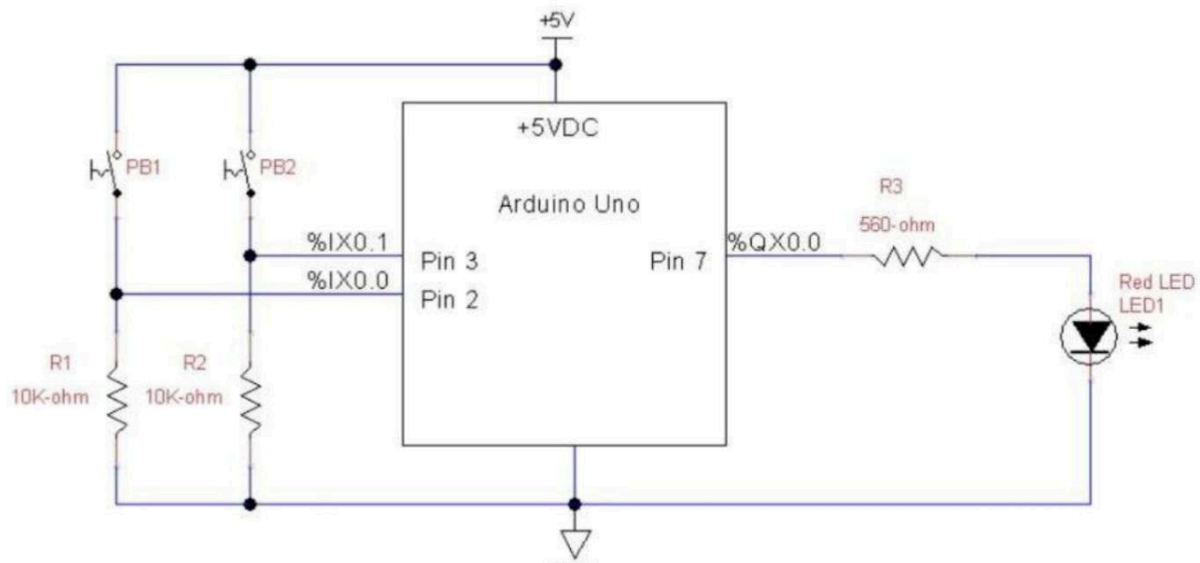


figure 2.0 start-stop control circuit

Mega / Due		
Digital In	A8, A9, A10, A11, A12, A13, A14, A15 22, 24, 26, 28, 30, 32, 34, 36 38, 40, 42, 44, 46, 48, 50, 52	%IX0.0 - %IX0.7 %IX1.0 - %IX1.7 %IX2.0 - %IX2.7
Digital Out	14, 15, 16, 17, 18, 19, 20, 21 23, 25, 27, 29, 31, 33, 35, 37 39, 41, 43, 45, 47, 49, 51, 53	%QX0.0 - %QX0.7 %QX1.0 - %QX1.7 %QX2.0 - %QX2.7
Analog In	A0, A1, A2, A3, A4, A5, A6, A7	%IW0 - %IW7
Analog Out	2, 3, 4, 5, 6, 7, 8, 9 10, 11, 12, 13	%QW0 - %QW7 %QW8 - %QW11

figure 3.0 Arduino Mega Addressing

METHODOLOGY

A. Ladder Diagram Development (OpenPLC Editor)

1. Project Initialization: The OpenPLC Editor software was launched, and a new project was started, selecting the language as LD (Ladder Diagram).
2. Variable Declaration: All necessary variables (UP_Start, DOWN_Stop, Latch, and LED) were created and declared as Local and BOOL (Boolean) type. The physical addresses (%IX0.0, %IX0.1, %QX0.0) were typed into the Location field for the I/O variables.
3. Ladder Diagram Construction: The logic for the Start-Stop Control Circuit was created using standard LD elements:
 - Rung 1 (Latching Logic): This rung contains the control logic for the internal Latch coil.
 - An NC contact for DOWN_Stop in series with an NO contact for UP_Start.
 - An NO contact for Latch is wired in parallel (seal-in) across the UP_Start contact.
 - The output is the Latch coil itself.
 - Rung 2 (Output Control): This rung controls the physical output.
 - An NO contact for Latch is wired in series.
 - The output is the LED coil (%QX0.0).

B. Simulation and Compilation

1. Software Compilation: The created ladder diagram was compiled within the OpenPLC Editor.

- Simulation: The program was simulated in the editor to verify the logic: ensuring the Latch and LED coils activate and hold with a momentary UP_Start signal and deactivate only with a DOWN_Stop signal.

C. Program Transfer to Arduino

- Hardware Connection: The Arduino Uno board was connected to the PC/laptop via a USB cable, and the corresponding COM port number was noted.
- Transfer Process: The compile icon was clicked, and the pop-up window was configured:
 - Board Type: Selected as *Arduino Mega*
 - COM Port: port number 7 was selected.
 - The Transfer to PLC button was pressed to upload the compiled program (OpenPLC Runtime) onto the Arduino.

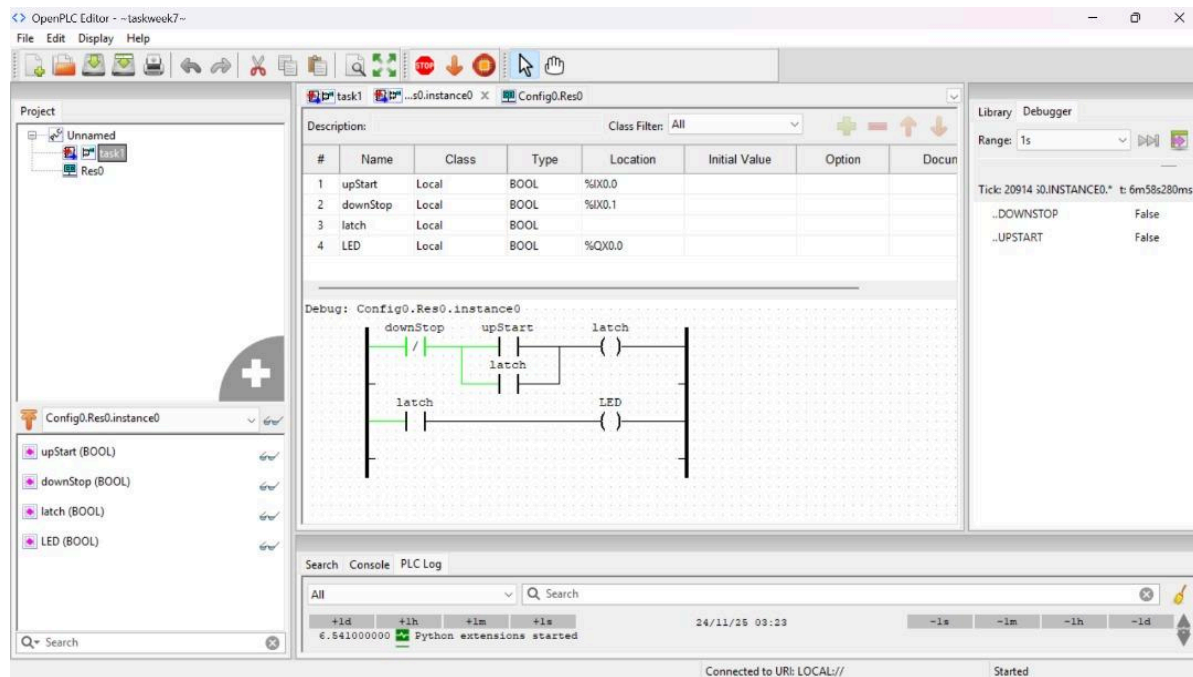


Figure 4.0 ladder diagram

DATA COLLECTION

step	state of UP_start(%IX0.0)	State of DOWN_Stop(%IX0.1)	observed output(LED %QX0.0)	observation
1	released (OFF)	released(OFF)	OFF	initial state
2	pressed-released	released(OFF)	ON	latching: LED activated and held ON
3	released(OFF)	released(OFF)	ON	circuit maintained by the latch bit
4	released(OFF)	pressed-released	OFF	unlatching: circuit broken and reset
5	released(OFF)	released(OFF)	OFF	system reset and ready for next start cycle

DATA ANALYSIS

The collected data is analyzed by comparing the observed output states to the expected logic derived from the Ladder Diagram for the Start-Stop Control Circuit.

1. Start Operation (Steps 2 and 3): When the UP_Start button is pressed, the current flows through the DOWN_Stop NC contact and the UP_Start NO contact to energize the internal Latch coil. Once energized, the auxiliary Latch NO contact (wired in parallel) closes, creating a "seal-in" path around the now-open UP_Start contact. This keeps the Latch coil energized and, consequently, the LED ON (Rung 2), satisfying the criteria for a latched circuit (Step 3).

2. Stop Operation (Step 4): The DOWN_Stop button is wired to a normally closed (NC) contact in the Ladder Diagram (figure 4.0). Pressing the physical button causes its corresponding input %IX0.1 to go HIGH (TRUE). Since the contact is negated (normally closed) in the LD, the current flow in Rung 1 is interrupted. This de-energizes the Latch coil, opening the seal-in contact and the LED control contact, causing the LED to turn OFF.

RESULTS

The experiment successfully achieved its objectives by creating, simulating, and deploying a functional Start-Stop Control Circuit using the OpenPLC Editor on an Arduino platform.

1. Successful Latching Logic: The Start-Stop control circuit operated as intended, validating the data presented in the State Transition Table:
 - a. Start/Latching: A momentary press of the UP_Start button (%IX0.0) successfully activated the internal Latch bit and the physical LED (%QX0.0), keeping the LED ON indefinitely until the Stop button was pressed (as shown in Step 3 of the data table).
 - b. Stop/Unlatching: A press of the DOWN_Stop button (%IX0.1) successfully interrupted the main circuit path, de-energizing the Latch bit and turning the LED OFF (as shown in Step 4 of the data table).
2. I/O Mapping Verification: The experiment verified the correct association between the OpenPLC logical variables and the Arduino's physical pins (e.g., UP_Start to Pin 2/%IX0.0, LED to Pin 7/%QX0.0), a crucial step for real-world deployment.

DISCUSSION

The experimental results demonstrated correct and reliable operation of a basic industrial Start-Stop control circuit using the OpenPLC Editor with an Arduino Mega. The recorded state transitions matched the intended Ladder Diagram (LD), confirming that the OpenPLC runtime successfully translated the high-level latching logic into physical I/O behaviour on the Mega's digital pins. The successful activation and sustained illumination of the LED after releasing the momentary Start button showed that the seal-in (latching) contact operated as designed. Similarly, the Stop button implemented as a normally closed (NC) contact correctly interrupted the latch and reset the circuit immediately when pressed.

These results imply that the Arduino Mega is capable of functioning as a low-cost, educational PLC platform. The Mega's larger number of I/O pins enhances its value for teaching and prototyping multi-input or multi-output control systems. The experiment confirms that fundamental PLC concepts such as latching, NO/NC behaviour, and state-based logic can be accurately replicated on microcontroller hardware through the OpenPLC ecosystem.

No major discrepancies appeared between expected and observed behaviour. Minor issues such as button bounce may exist in theory, but the relatively slow scan cycle and simple digital logic minimized their impact. The system behaved consistently with standard PLC ladder logic principles.

CONCLUSION

This experiment had several limitations based on the use of an Arduino Mega as the control platform. Although the Mega provides significantly more I/O compared to common educational microcontrollers, its scan cycle remains slower than industrial-grade PLCs, making it unsuitable for high-speed or time-critical automation. Additionally, the Mega's 5 VDC I/O cannot directly interface with industrial loads without appropriate isolation hardware such as relays or optocouplers. The breadboard wiring used in the experiment also introduces potential electrical noise and instability compared to industrial wiring standards.

Despite these limitations, the core findings were clear and consistent. The Arduino Mega, running the OpenPLC runtime, successfully implemented the Start-Stop latching circuit, including correct latching, unlatching, and I/O mapping. The hardware behaviour matched the expected ladder logic precisely. These results directly support the hypothesis: a momentary press

of the UP_Start push button successfully latched the LED ON, and the LED remained ON until the DOWN_Stop push button was pressed, which correctly unlatched the circuit and turned the LED OFF.

The significance of these findings lies in demonstrating that the Arduino Mega can function as an effective, low-cost platform for learning and prototyping fundamental PLC concepts. The experiment validates that core industrial control principles such as latching logic, the use of NO/NC contacts, and state-based outputs can be accurately replicated on accessible embedded hardware through the OpenPLC ecosystem. The concepts demonstrated in this experiment apply directly to broader industrial automation scenarios, including pump control, motor starters, conveyor systems, and safety interlocks. Start-Stop circuits form the foundation of many machine control applications, and the Arduino Mega provides enough I/O to expand into larger or multi-device automation tasks in future labs.

RECOMMENDATIONS

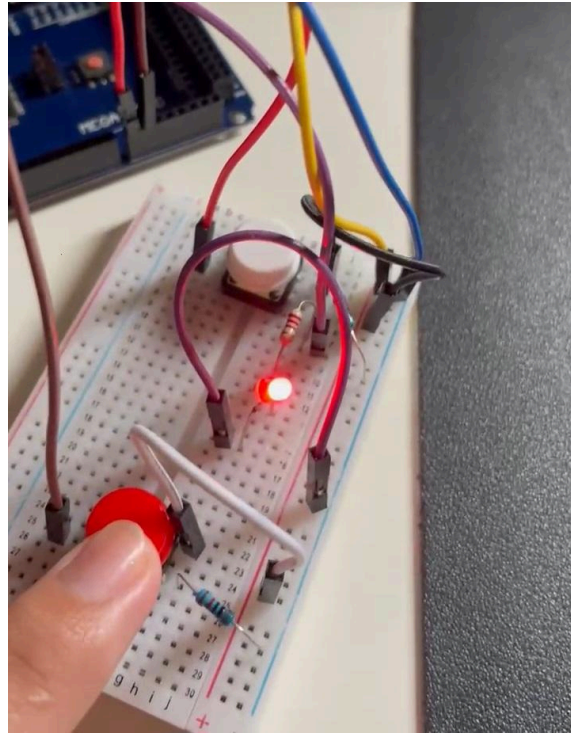
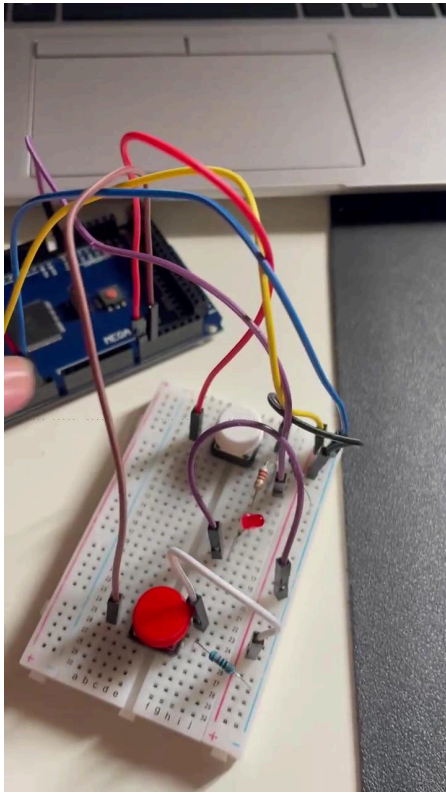
For future improvements, replacing the LED with a relay module or motor driver is recommended. This would allow the Arduino Mega to control an actual actuator, providing a more realistic demonstration of electromechanical outputs. Introducing a simulated fault input such as an overload or emergency-stop contact would also enhance the learning experience by showing how NC safety logic overrides normal Start commands.

Several important lessons emerged from the experiment. Accurate I/O mapping (e.g., assigning correct %IX and %QX addresses to Mega pins) is essential to ensure correct hardware-software interaction. Students must also understand how physical momentary pushbuttons translate into NO or NC contacts in ladder logic, as this skill is foundational to designing safe and reliable control systems.

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APPENDICES



ACKNOWLEDGEMENTS

Special thanks to Zulkifli Bin Zainal Abidin & Wahju Sediono for their guidance and support during this experiment. We would also like to extend our gratitude to our instructors, peers, and the laboratory staff for their valuable insights and assistance throughout the experiment. Their feedback and encouragement have greatly contributed to the success of this project. Finally, we appreciate the resources and learning materials provided, which enabled us to understand and implement serial communication effectively.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been undertaken or done by unspecified sources or persons. We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report. We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

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